Assessment criteria for Practice 1

Cloud Computing Services
Architectures and Infrastructures for Artificial Intelligence

This assignment assesses the development of a detection system for smart lighting, its operation locally and on Azure, and the analysis of its performance with multiple cameras. The base mark is out of 10 points, with up to 2 additional points for advanced work.

Grade Assignment

Assessment block	Tasks	Weight	Maximum
Part 1 · Local processing	1.1 to 1.3	10 %	1 point
Part 2 · Azure virtual machine	2.1 to 2.5	30 %	3 points
Part 3 · Serverless and Azure AI	3.1 to 3.6	30 %	3 points
Part 4 · Multi-camera management	4.1 to 4.3	20 %	2 points
Project report	All parts	10 %	1 point
Total base mark		100 %	10 points
Optional advanced work	Extensions	Additional	Up to +2.0

Tasks are assessed together within each block. Full marks require evidence that all listed criteria have been met, supported by results and explanations that allow the work to be verified. Shortcomings in implementation, analysis or documentation reduce the mark for the relevant block.

Part 1: Local processing

Maximum: 1.0 point

Task	Assessment criteria
1.1	Select a fixed public webcam showing a street with people or vehicles.
1.2	Capture one frame locally every 5 seconds, at a resolution sufficient to detect the relevant objects.
1.3	Count people using YOLOv8, save the count and date in a CSV file, and plot the number of people per frame over time.

Evidence: webcam images and detection results, a sample of the CSV file, and the time-series graph.

Part 2: Azure virtual machine

Maximum: 3.0 points

Task	Assessment criteria
2.1	Create and configure an Azure virtual machine with the required dependencies.
2.2	Send frames captured on the local machine to the virtual machine.
2.3	Run person detection and counting with YOLOv8 on the virtual machine and save the results in a CSV file.
2.4	Publish a histogram and other results on a website hosted on the virtual machine and accessible through a public IP address.
2.5	Measure and analyse detection effectiveness, memory and CPU usage, bandwidth consumption, monetary cost and response time.

Evidence: virtual machine configuration, image reception and processing, CSV output, the results website, and performance measurements. Explain what the metrics show and use tables or graphs to support their interpretation.

Part 3: Serverless and Azure AI

Maximum: 3.0 points

Task	Assessment criteria
3.1	Create a static website in an Azure storage account to display the results.
3.2	Upload images continuously to a container separate from the website container.
3.3	Configure a function with a blob trigger that runs only when images are uploaded to the input container.
3.4	Create and configure the Azure AI service required for image processing.
3.5	Integrate the function with Azure AI to count people and draw a box around each detected person.
3.6	Measure and analyse detection effectiveness, monetary cost and response time for the serverless application.

Evidence: configured containers and trigger, function executions, results displayed on the website, and performance analysis. Clearly explain how the components connect and operate. In the performance analysis, state the test conditions, how measurements were obtained, and their units.

Part 4: Multi-camera management

Maximum: 2.0 points

Task	Assessment criteria
4.1	Gradually increase the number of cameras and send one frame every 5 seconds per camera.
4.2	Test the virtual machine with multiple cameras and identify its processing limit. Measure memory and CPU usage, bandwidth consumption, cost and response time.
4.3	Test the serverless application with multiple cameras and identify its processing limit. Measure detection effectiveness, cost and response time.

Provide graphs or tables relating the number of cameras to performance and allowing both architectures to be compared. Justify the observed limits and explain the behaviour of the implementation.

Project report

Maximum: 1.0 point

The report must make the work carried out in every part understandable and verifiable. The report should contain the following sections:

- An introduction explaining the problem and the objectives of the assignment.
- Solution of each task, ordered by task number:
  - A structured explanation of the steps followed, problems encountered, and design decisions made.
  - Relevant results and screenshots, with metrics and graphs explained in the text.
- Conclusions supported by the work carried out, and references to the resources used.

Clear, coherent and complete writing will also be assessed.

Submission

Work in groups of 2 or 3 students. Submit the report as a PDF through the Moodle assignment.